Cosmos DB: Microsoft's New Globally Distributed and Horizontally Scalable Database Service

Data is gold in today's world. But it has no value unless it is processed, manipulated and used to derive useful insights. Databases enable the handling of data, and there are many out there to choose from. Cosmos DB has a lot going for it, as readers will discover in this article.

(Credit: https://codemilltech.com)

Cosmos DB is the next generation of Azure DB and an enhanced version of Document DB. Azure Cosmos DB is Microsoft's globally distributed, multi-modal database, which scales at the click of a button. Because it's multi-modal, it supports document, table and graph values together in a single database. Cosmos DB (formerly known as Document DB) is designed to store semi-structured data such as documents, typically in JSON or XML format. Unlike traditional relational databases, the schema for each non-relational (NoSQL) document can vary, giving developers, database administrators and IT professionals more flexibility in organising and storing application data and reducing the storage required for optional values.

Cosmos DB is a planet scale database. It is a good choice for any serverless application that needs low order-of-millisecond response times, and has to scale rapidly and globally. It is more transparent to our application and *config* does not need to change.

How it was created

Microsoft Cosmos DB isn't entirely new — it grew out of a Microsoft development initiative called Project Florence that began in 2010. Project Florence is a speculative glimpse into our future. It envisages both our natural and digital worlds co-existing in harmony through enhanced communication.

- It was first commercialised in 2015 with the release of a NoSQL database called Azure Document DB.
- Cosmos DB was introduced in 2017.
- It expands on Document DB by adding multi-modal support, global distribution capabilities and relational-like guarantees for latency, throughput, consistency and availability.

Defining Cosmos DB

Cosmos DB is a database service that is globally distributed. It allows for the management of data even if it is stored in data centres that are scattered throughout the world. It provides the tools needed to scale both global distribution patterns and computational resources that are provided by Microsoft Azure.

Reasons for using Cosmos DB

- It has no data scheme and is schema-free. It indexes all the data without requiring schema and index management.
- It is also a multi-modal, natively supporting document, key-value, graph and column-family data model.
- It is the industry's first globally distributed, horizontally scalable, multi-modal database service. Azure Cosmos DB guarantees single-digit-millisecond latencies at the 99th percentile anywhere in the world. It offers multiple well-defined consistency models to fine-tune performance and guarantees high availability (HA).
- There's no need to worry about instances, servers, CPU or memory. Just select the throughput, the required storage and create collections. Cosmos DB works based only on throughputs. It is integrated with Azure functions. It is a serverless event-driven solution.
- APIs and access methods: Cosmos DB comes with the Document DB API, Graph API (Gremlin), Mongo DB API, RESTful HTTP API and the Table API. This gives more flexibility to the developer.